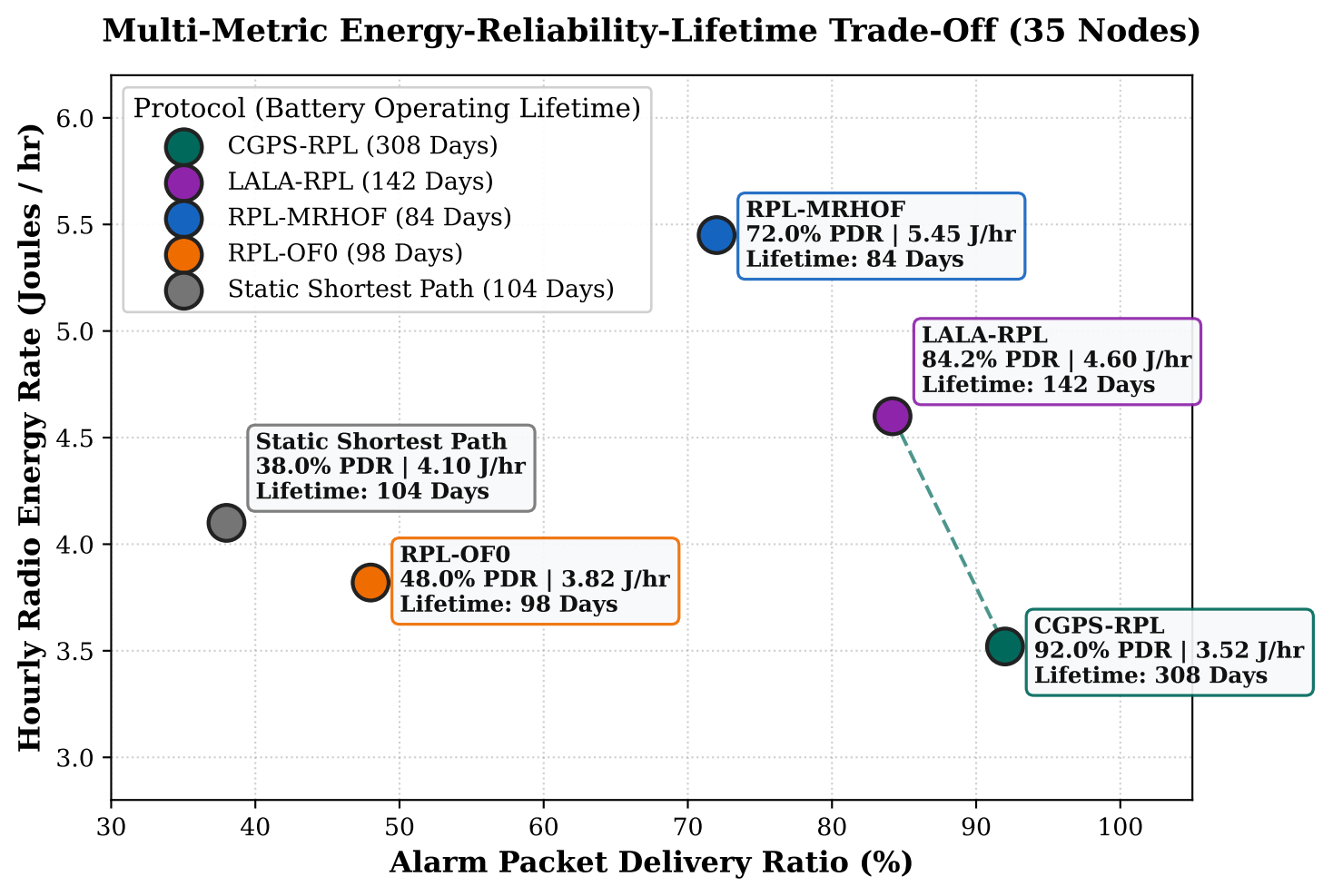
Multi-Metric Energy-Reliability-Lifetime Trade-Off (35 Nodes)
Protocol (Battery Operating Lifetime)
CGPS-RPL (308 Days)
LALA-RPL (142 Days)
RPL-MRHOF (84 Days)
RPL-OF0 (98 Days)
Static Shortest Path (104 Days)
RPL-MRHOF
72.0% PDR | 5.45 J/hr
Lifetime: 84 Days
LALA-RPL
84.2% PDR | 4.60 J/hr
Lifetime: 142 Days
Static Shortest Path
38.0% PDR | 4.10 J/hr
Lifetime: 104 Days
RPL-OF0
48.0% PDR | 3.82 J/hr
Lifetime: 98 Days
CGPS-RPL
92.0% PDR | 3.52 J/hr
Lifetime: 308 Days
Hourly Radio Energy Rate (Joules / hr)
6.0
5.5
5.0
4.5
4.0
3.5
3.0
30
40
50
60
70
80
90
100
Alarm Packet Delivery Ratio (%)